

Nguyen Thanh Duy Tan

AI Engineer Fresher

District 7, Ho Chi Minh City | +84 339 900 276 | tanntd.2005@gmail.com
github.com/TanNguyen234 | linkedin.com/in/tannguyen234 | nguyentan234.vercel.app

PROFILE

Early-career AI engineer and Information Technology student focused on applied AI systems, retrieval/evaluation, and Python backend engineering. Built and deployed RAG, multi-agent LLM, and ML inference services with reproducible evaluation, automated testing, Docker, and CI/CD.

EDUCATION

Ho Chi Minh City University of Natural Resources and Environment

Oct 2023 - Present
Ho Chi Minh City

Major: Information Technology | GPA: 3.0/4.0
Academic Encouragement Scholarship, Academic Year 2024-2025

TECHNICAL SKILLS

Applied AI: RAG, LangGraph, LangChain, multi-agent workflows, hybrid retrieval, LLM evaluation

Machine Learning: Scikit-learn, XGBoost, Pandas, NumPy, imbalanced classification, EDA

Backend & MLOps: Python, FastAPI, REST APIs, SSE, Docker, Pytest, GitHub Actions

Data & Cloud: Qdrant, MongoDB/Motor, SQLite/FTS5, Supabase, Google Vertex AI

SELECTED PROJECTS

VietLex - Vietnamese Legal RAG & Evidence Workflow

07/2026 - Present

[GitHub](#) | [Live Demo](#)

- Built a Vietnamese legal RAG/research system backed by a pinned 518,255-document corpus; the current v3 runtime indexes 141,798 structural points across 14,962 audited documents using Vertex embeddings, Qdrant dense-sparse retrieval, and fail-closed routing.
- Built versioned Golden-50 retrieval/generation evaluation with evidence provenance and regression tracking; the current v3 ranking uses RRF+DBSF, and the verified subset records 53/53 Document Recall@3.
- Shipped citation-backed Q&A, legal-document search, research workspaces, source reading/OCR, report export, and feedback/regression workflows; maintained 1,200+ automated tests plus scoped live verification.

Stack: Python, FastAPI, Vertex AI, Qdrant, Supabase, SQLite/FTS5, MongoDB, NeMo Guardrails, Ragas, Pytest, Docker

CV AI Evaluation System - Multi-Agent LLM Application

03/2026 - 05/2026

[GitHub](#) | [Live Demo](#)

- Built a LangGraph pipeline for PDF extraction, candidate profiling, market enrichment, parallel fan-out across six specialized evaluators, deterministic validation, optional JD analysis, and meta-evaluation.
- Added configurable Gemini/Qwen providers with retry/backoff for transient LLM failures; served jobs through async FastAPI, SSE progress streaming, and MongoDB/Motor persistence.
- Implemented CV validation, per-IP limits, unit/integration tests, and Docker deployment on Hugging Face Spaces.

Stack: Python, FastAPI, LangGraph, LangChain, Gemini, Qwen, Tavily, MongoDB/Motor, PyMuPDF4LLM, Docker

Fraud Detection API - ML Inference & CI/CD Pipeline

05/2026 - 06/2026

[GitHub](#)

- Trained an XGBoost classifier on the ULB credit-card dataset (~0.17% fraud) using duplicate removal, hour-of-day transformation, RobustScaler, stratified splitting, and scale_pos_weight.
- Evaluated with PR-AUC and class-wise metrics, then served versioned FastAPI/Pydantic endpoints for single and batch inference of up to 100 transactions.
- Built a GitHub Actions pipeline that runs Ruff and Pytest before building and pushing the Docker image to Docker Hub on main.

Stack: Python, XGBoost, Scikit-learn, Pandas, FastAPI, Pydantic, Pytest, Docker, GitHub Actions